

Container Orchestration Tool – Kubernetes

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L2 - Scale-up and Scale-Down the Pods Deployed

Step :1 login into AWS console and create two ec2 instances

The screenshot displays the AWS Management Console interface for the 'us-east-1' region. The 'Instances' page shows two EC2 instances: 'master-server' and 'worker-server'. Both instances are in the 'Running' state. The 'master-server' has an Instance ID of 'i-04002c4ef19509d70' and the 'worker-server' has an Instance ID of 'i-05ab4719b9951a3f5'. Both are using the 'c7i-flex.large' instance type. The console also shows a 'Monitoring' section with various metrics like CPU utilization, Network in/out, and Network packets in/out, all of which currently show 'No data available'.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
master-server	i-04002c4ef19509d70	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b	ec2-3-82-121-146.com...
worker-server	i-05ab4719b9951a3f5	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b	ec2-35-173-177-24.co...

Step :2 This is master node Terminal

The screenshot shows a terminal window for the master node i-04002c4ef19509d70. The terminal is running the following commands and output:

```
[root@ip-172-31-82-2 ec2-user]# kubeadm token create --print-join-command
kubeadm join 172.31.82.2:6443 --token 3y4te6.azjb5q2c16lucxlc --discovery-token-ca-cert-hash sha256:03f82099568a11b07caf626ae81470a19cb116de2434dd292b53eaf37663d74
[root@ip-172-31-82-2 ec2-user]# kubectl get nodes
NAME                                STATUS    ROLES    AGE     VERSION
ip-172-31-82-2.ec2.internal          Ready     control-plane  2m20s   v1.29.15
ip-172-31-93-157.ec2.internal        Ready     <none>      26s     v1.29.15
[root@ip-172-31-82-2 ec2-user]#
```

The terminal window is titled "i-04002c4ef19509d70 (master-server)" and shows the public IP as 3.82.121.146 and private IP as 172.31.82.2. The AWS console header shows the region as "United States (N. Virginia)" and the user as "Rahul (8106-4823-6072)".

Step :3 This is worker node Terminal

The screenshot shows a terminal window for the worker node i-05ab4719bf951a3f5. The terminal is running the following commands and output:

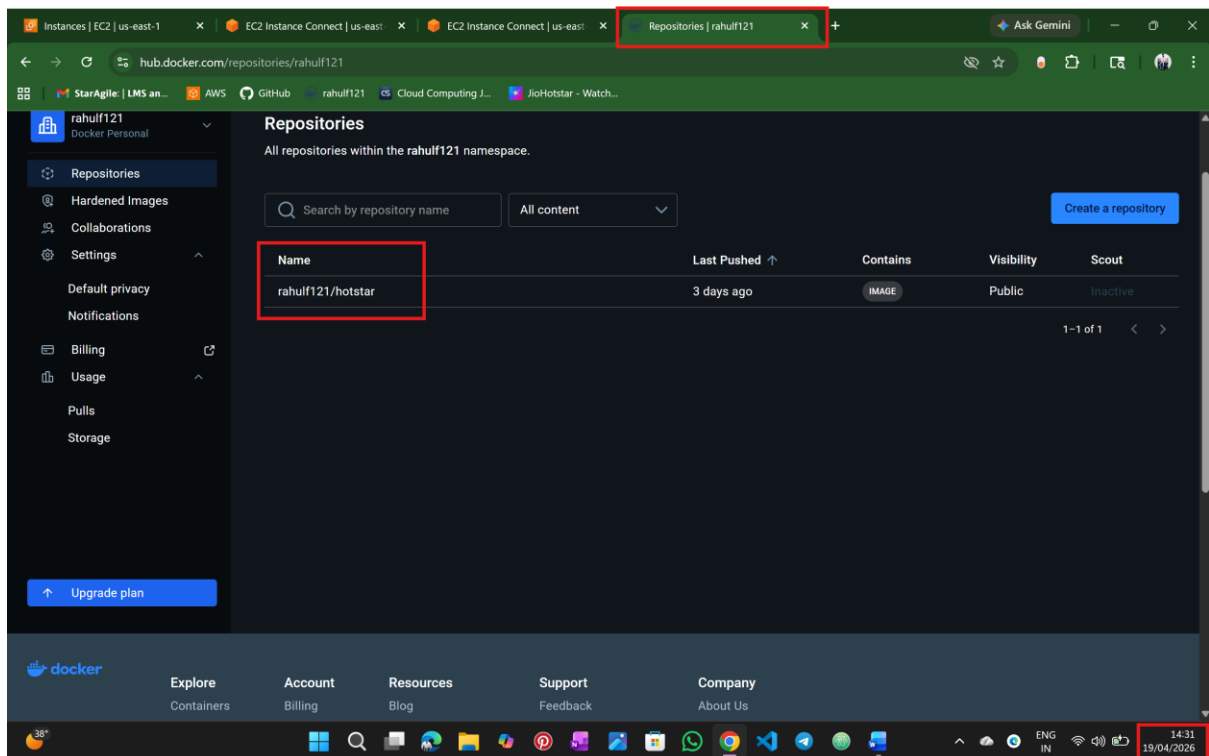
```
[root@ip-172-31-93-157 ec2-user]# kubeadm join 172.31.82.2:6443 --token 3y4te6.azjb5q2c16lucxlc --discovery-token-ca-cert-hash sha256:03f82099568a11b07caf626ae81470a19cb116de2434dd292b53eaf37663d74
[WARNING] Running pre-flight checks
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap...

This node has joined the cluster:
* Certificate signing request was sent to apiservert and a response was received.
* The Kubelet was informed of the new secure connection details.

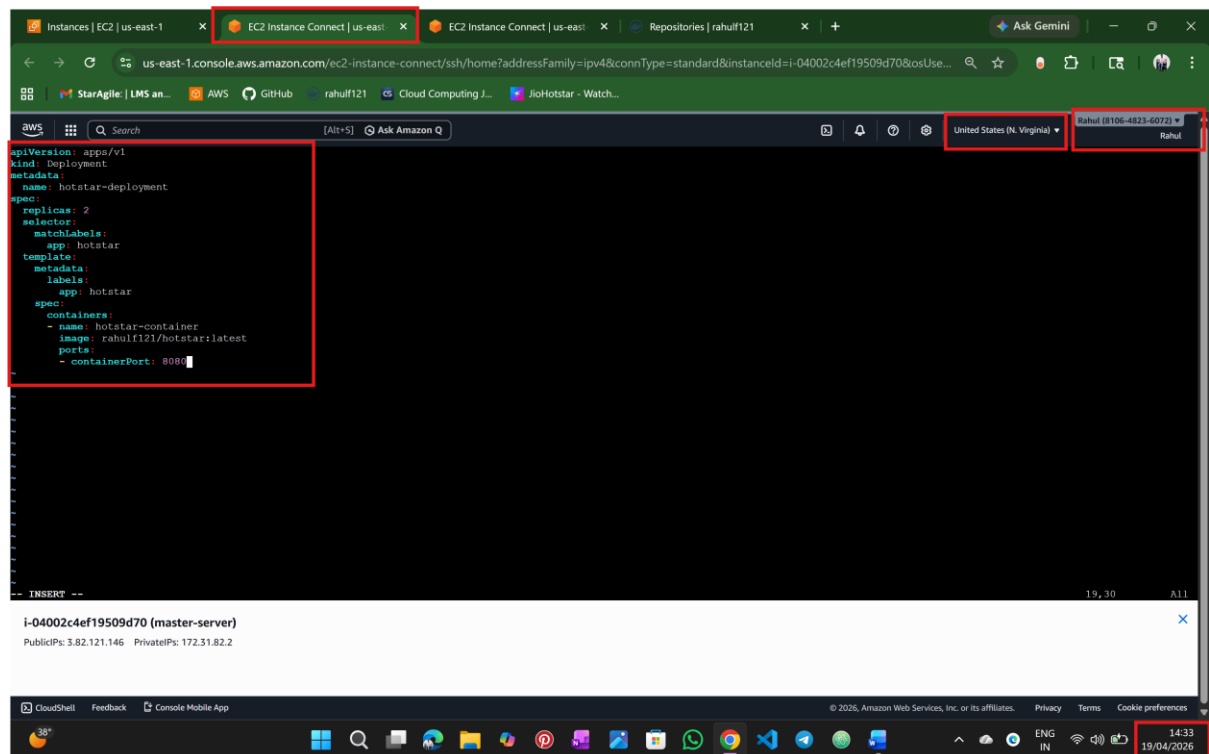
Run 'kubectl get nodes' on the control-plane to see this node join the cluster.
[root@ip-172-31-93-157 ec2-user]#
```

The terminal window is titled "i-05ab4719bf951a3f5 (worker-server)" and shows the public IP as 35.173.177.24 and private IP as 172.31.93.157. The AWS console header shows the region as "United States (N. Virginia)" and the user as "Rahul (8106-4823-6072)".

Step :4 I have already docker image in my docker hub. If, not push it.



Step :5 This is the deployment.yaml



Step :5 This is the service.yaml

The screenshot shows the AWS Management Console for an EC2 instance. The browser tabs include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'Repositories | rahul121'. The console shows the 'service.yaml' file with the following content:

```
apiVersion: v1
kind: Service
metadata:
  name: hotstar-service
spec:
  type: NodePort
  selector:
    app: hotstar
  ports:
    - port: 80
      targetPort: 8000
      nodePort: 30007
```

The console also shows the instance details for 'i-04002c4ef19509d70 (master-server)' with PublicIPs: 3.82.121.146 and PrivateIPs: 172.31.82.2. The bottom of the screen shows the Windows taskbar with the date and time '14:34 19/04/2026'.

Step :6 Now apply and verify

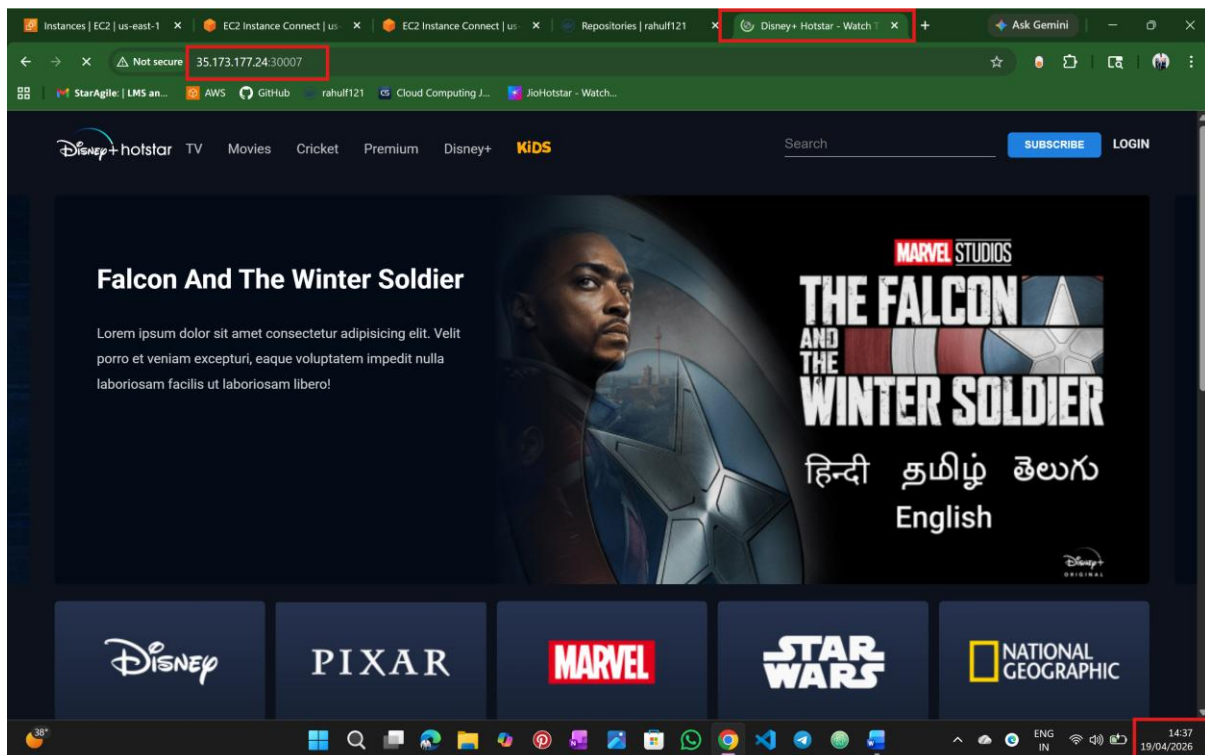
→access the website

The screenshot shows the AWS Management Console for an EC2 instance. The browser tabs include 'Instances | EC2 | us-east-1', 'EC2 Instance Connect | us-east-1', and 'Repositories | rahul121'. The console shows the terminal output for applying and verifying the service:

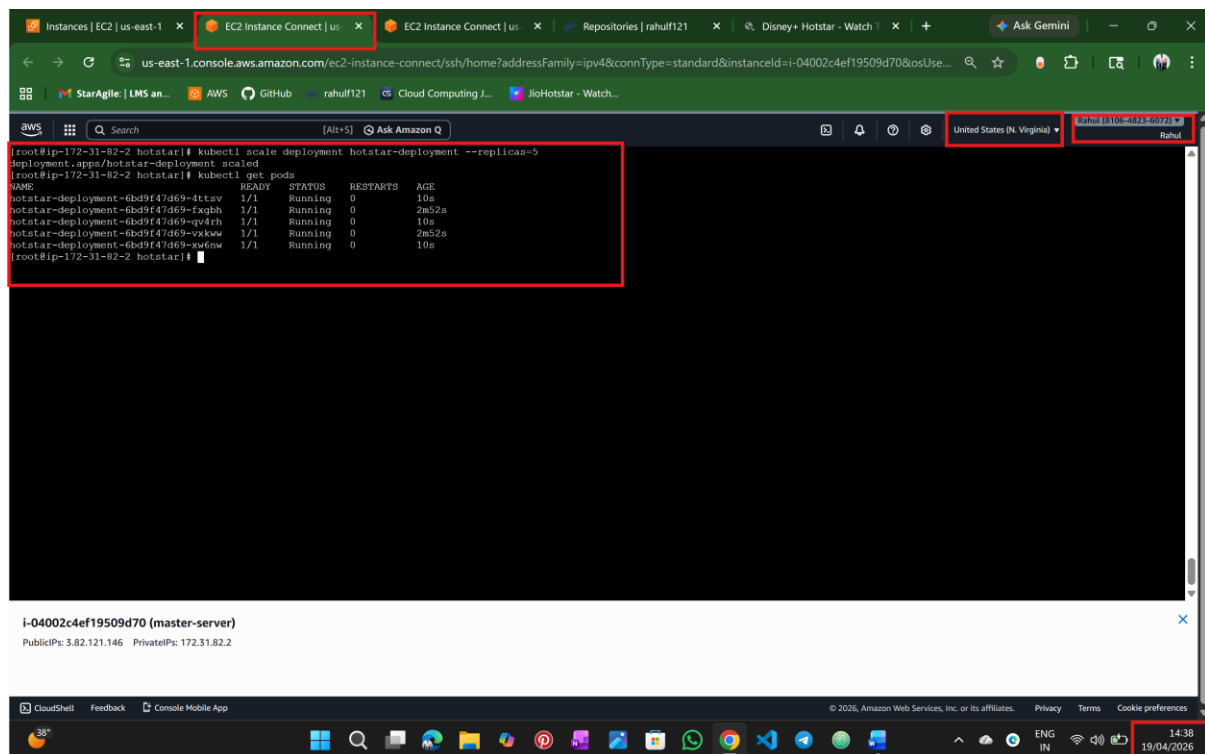
```
[root@ip-172-31-82-2 ec2-user]# kubectl token create --print-join-command
kubectl join 172.31.82.2:6443 --token 3y4te6.azjh5q2c16lucxlc --discovery-token-ca-cert-hash sha256:03f82099568a11b07caf626ae81470a19cb116de2434dd292b53eaa3f37663d74
[root@ip-172-31-82-2 ec2-user]# kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-172-31-82-2.ec2.internal          Ready    control-plane  2m20s  v1.29.15
ip-172-31-93-157.ec2.internal         Ready    <none>      26s    v1.29.15
[root@ip-172-31-82-2 ec2-user]# mkdir hotstar
[root@ip-172-31-82-2 ec2-user]# cd hotstar/
[root@ip-172-31-82-2 hotstar]# ls
[root@ip-172-31-82-2 hotstar]# vim deployment.yaml
[root@ip-172-31-82-2 hotstar]# vim service.yaml
[root@ip-172-31-82-2 hotstar]# kubectl apply -f deployment.yaml
deployment.apps/hotstar-deployment created
[root@ip-172-31-82-2 hotstar]# kubectl apply -f service.yaml
service/hotstar-service created
[root@ip-172-31-82-2 hotstar]# kubectl get pods
NAME                                READY    STATUS    RESTARTS   AGE
hotstar-deployment-6bd9f47d69-fxgbh 1/1      Running   0           17s
hotstar-deployment-6bd9f47d69-vxkew 1/1      Running   0           17s
[root@ip-172-31-82-2 hotstar]# kubectl get svc
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
hotstar-service                     NodePort    10.98.6.8      <none>          80:30007/TCP  9s
kubernetes                          ClusterIP   10.96.0.1      <none>          443/TCP     12m
[root@ip-172-31-82-2 hotstar]#
```

The console also shows the instance details for 'i-04002c4ef19509d70 (master-server)' with PublicIPs: 3.82.121.146 and PrivateIPs: 172.31.82.2. The bottom of the screen shows the Windows taskbar with the date and time '14:36 19/04/2026'.

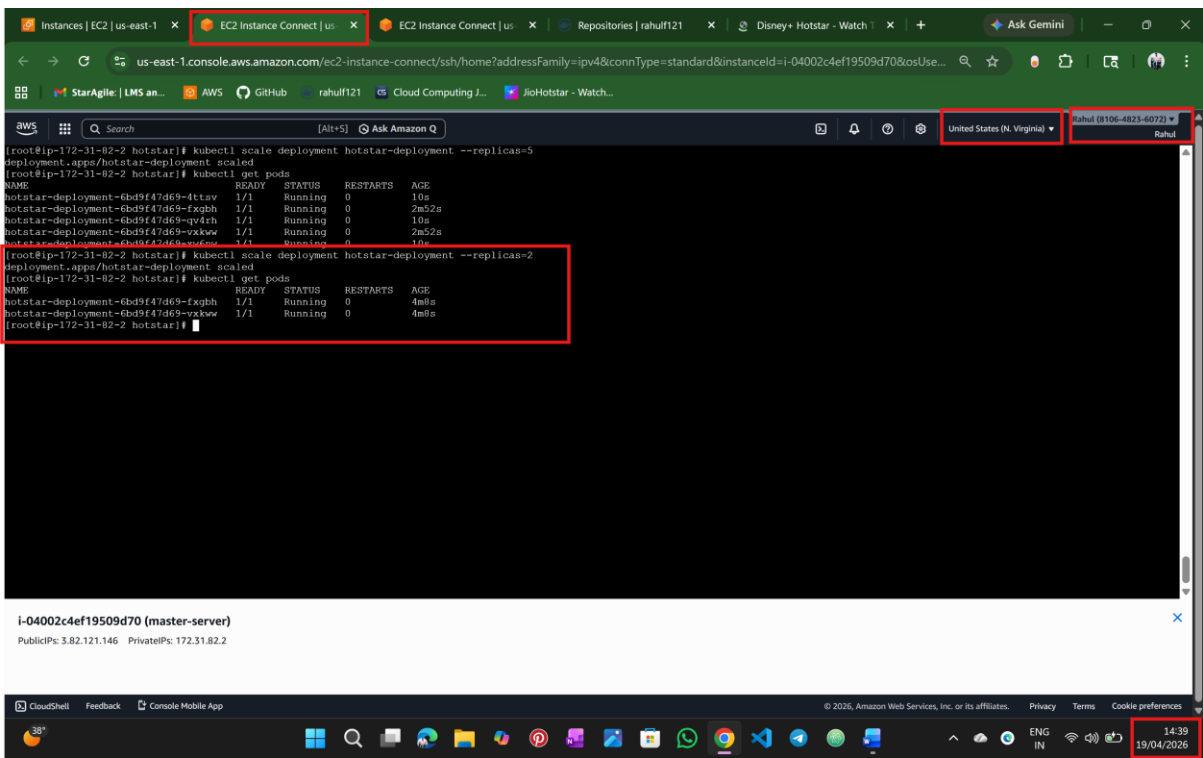
Step: 7 access the website with worker <ip : port>



Step :8 scale up



Step :9 scale down



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Nalathati Rahul